Databricks basics

by Nicola Orecchini, Mar 2026

Databricks is a multi-cloud[1] data and analytics platform providing a managed environment to build enterprise data and analytics use-cases

databricks

Optimized Spark runtime
An enhanced Spark version containing performance optimizations, pre-installed libraries, integrations with cloud storage

Delta lake
A layer on top of storage, adding to it features such as ACID properties, time travel, upsert

Cluster manager
Create clusters for Spark to run on, including VM provisioning, autoscaling, monitoring, logging

Unity catalog
A governance layer on top of storage that allows to manage data lineage, access, catalog, auditing

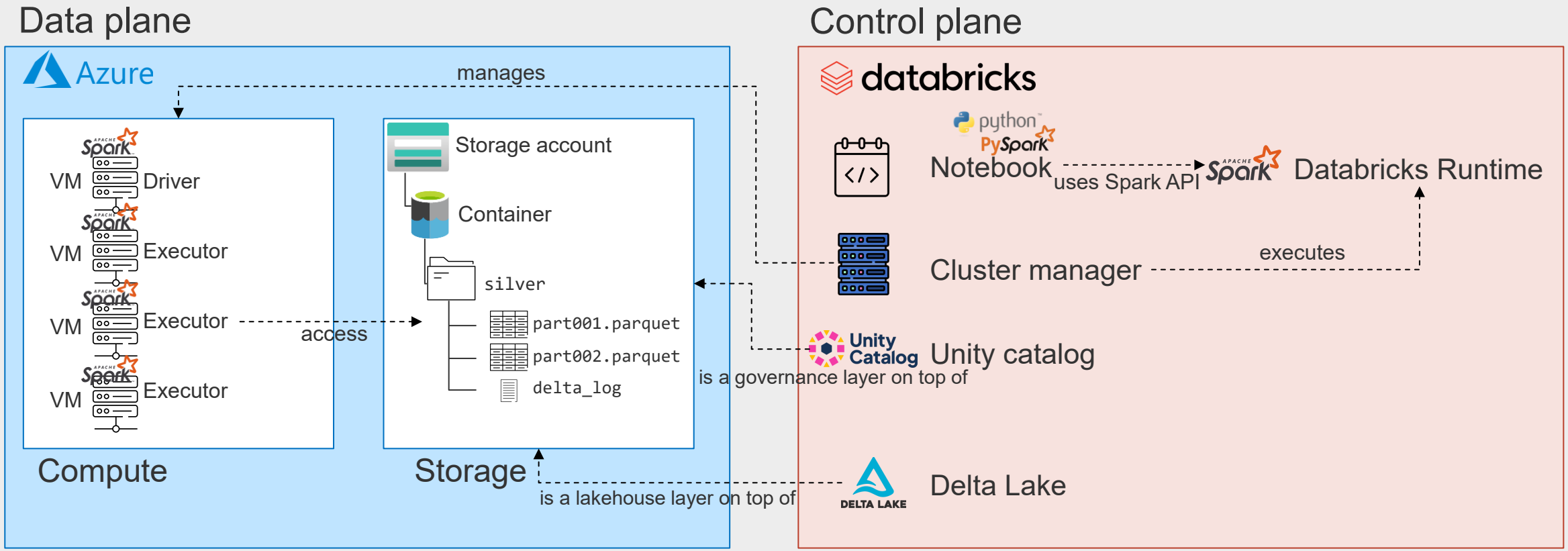
Collaborative workspace
Notebooks to develop data/ML pipelines/workloads, integration with git for version control, libraries management

Databricks is built around Apache Spark, but while the latter is open source, Databricks is its industrialized version of it

Databricks is like an operating system for data and AI: without it, companies would need to manage Spark clusters, autoscaling, job scheduling, monitoring, security, data catalog, and other, all through dozens of open source tools

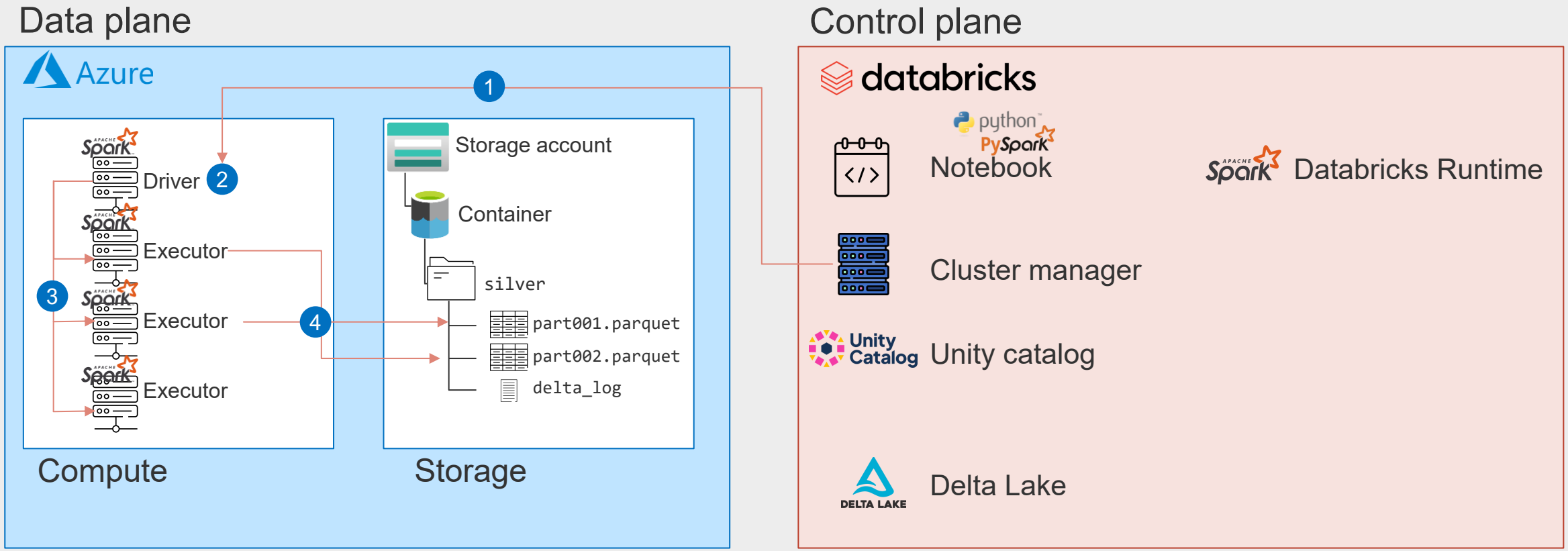
1. Databricks exists on top of major cloud providers (Amazon Web Services, Microsoft Azure, and Google Cloud)

Spark doesn't provide resources, though: it runs on top of a Cloud Services Provider



VM = Virtual Machine, i.e. a simulated computer inside a physical server (including RAM, CPU, disk, OS, etc. – like a regular computer)

When code is executed on Databricks, the workload is transfered to the underlying could infrastructure



1 The driver receives the command

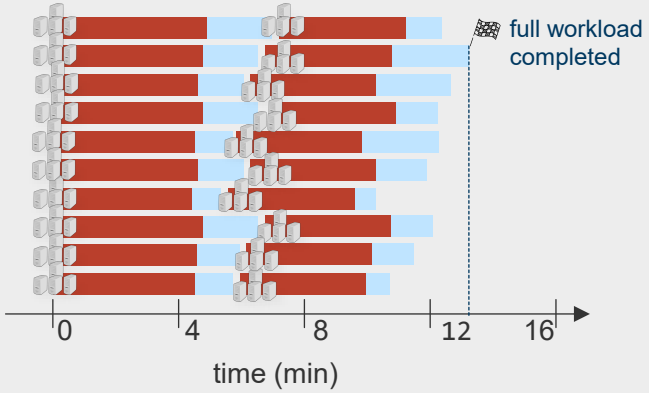
2 Spark builds the execution plan and divides the workload in parallel tasks

3 Tasks are sent to the workers

4 Workers read/write from/to the storage

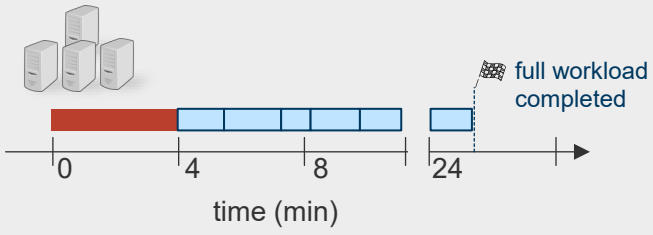
Use case: 20 independent pipelines

ForEach loop with parallelism=10 clusters



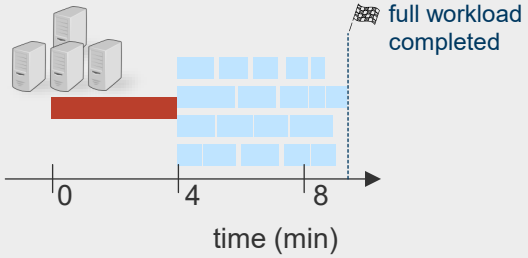
Time to finish: 20*[4+(~1)]/10≈10 min

One single cluster, sequential



Time to finish: 4+20*(~1)≈24 min

One single cluster, ThreadPool(4)



Time to finish: 4+20/4*(~1)≈9 min

# clusters running at a time	10	1	1
# driver per cluster	1	1	1
# worker per cluster (autoscaling interval)	1-3	1-3	1-3
# VMs running at a time	10*(3+1)=40	1*(3+1)=4	1*(3+1)=4
Tot running time	10 min	24 min	9 min

Server (Virtual Machine)

Cluster

Cluster startup time

Pipeline processing time